COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.6

This worksheet contains a combination of *formative activities* (which contribute towards your learning) and *summative activities* (which you will complete and submit to be assessed as part of your portfolio).

- Exercises marked with red borders are summative and must be submitted in Gradescope.
- Other exercises may be covered in tutorial sessions.
- Exercises marked by (*) are advanced, and may take some time to complete.
- Exercises marked by (*E*) are exam-style questions. You should expect to see questions like this in the exam.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided and submit your portfolio evidence on PebblePad before the deadline.
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff.
3. **Integrity:** You are responsible for ensuring that the evidence you submit as part of your portfolio is **entirely your own work**. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in tutorials and the Math Lunch Hour. Individual support is available via office hours.

Expected completion date: No due date

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment. Within this assessment you may use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must not use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning outcomes

On completion of this activity sheet you will have:

1. understood number representations in multiple bases,
2. efficiently carried out addition, multiplication, and exponentiation in modular arithmetic,
3. explained residue classes and modular arithmetic,
4. applied algorithms to compute the greatest common divisor.

Tutorials on Wednesday and Thursday will cover material from Week 5, Lecture 1. This material is in exercises 1–8. Tutorials on Monday and Tuesday will cover material from Week 5, Lecture 2. This material is in exercises 9–14.

Questions

1.
   (a) Express $(D89EA)_{16}$ in decimal notation.
   (b) Express $(704361)_8$ in decimal notation.
   (c) Express $(230)_{10}$ in binary notation.
   (d) Express $(1002)_{10}$ in octal notation.

2. Let $n > 1$ be an integer. How many numbers can be represented with *exactly* n digits chosen from

$$\{0, 1, 2, \ldots, 9, A, B, C, D, E, F\}?$$

   Remember that the first digit cannot be zero.

3. Use Algorithm 5 on page 93 of the lecture notes to complete the following tasks. Assume that the base b is equal to 10.
   (a) Find the quotient q and remainder r when $m = 723$ and $n = 32$.
   (b) Find the quotient q and remainder r when $m = 1041$ and $n = 56$.

4. Use Algorithm 5 on page 93 of the lecture notes to complete the following tasks. Assume that the base b is equal to 8.
   (a) Find the quotient q and remainder r when $m = (307)_8$ and $n = (12)_8$.
   (b) Find the quotient q and remainder r when $m = (4171)_8$ and $n = (635)_8$.

5. Make the following basis conversions by repeatedly finding quotients and remainders. (Follow the example on page 94 of the lecture notes.)
   (a) Express $(203)_{10}$ in octal representation.
   (b) Express $(114)_{10}$ in binary representation.
   (c) Express $(1051)_{10}$ in hexadecimal representation.

6. Use Algorithm 6 on page 96 of the lecture notes to calculate the following values.
   (a) $x^y \bmod 7$ when $x = (10)_2$ and $y = (1001011)_2$ (so $n = 7$).
   (b) $x^y \bmod 5$ when $x = (11)_2$ and $y = (101101)_2$ (so $n = 5$).
7. Carry out the following modular arithmetic operations
   (a) $(11 + 12) \bmod 19$
   (b) $(7 \cdot 4) \bmod 16$
   (c) $2^4 \bmod 3$
   (d) $(8 \cdot 6) \bmod 12$
8. Say that the sequence $r_0, r_1, r_2, \ldots$ is defined by $r_0 = 273$, $r_1 = 153$, and r_i is the remainder r_{i-2} modulo r_{i-1}.
   (a) What is r_5?
   (b) What is the smallest integer i such that $r_i = 0$?
   (c) What is $\gcd(273, 153)$?
   (d) Give integers x and y such that $273x + 153y = \gcd(273, 153)$.
   (e) Does 153 have a multiplicative inverse modulo 273? If so, what is it?
   (f) What is the least common multiple of 273 and 153?
9. Say that the sequence $r_0, r_1, r_2, \ldots$ is defined by $r_0 = 310$, $r_1 = 203$, and r_i is the remainder r_{i-2} modulo r_{i-1}.
   (a) What is r_3?
   (b) What is the smallest integer i such that $r_i = 0$?
   (c) What is $\gcd(310, 203)$?
   (d) Give integers x and y such that $310x + 203y = \gcd(310, 203)$.
   (e) Does 203 have a multiplicative inverse modulo 310? If so, what is it?
   (f) What is the least common multiple of 310 and 203?
10. Use Algorithm 9 on page 101 of the lecture notes to find $\gcd(7, 20)$ and to find integers x and y such that $7x + 20y = \gcd(7, 20)$.
11. Use Algorithm 9 on page 101 of the lecture notes to find $\gcd(36, 28)$ and to find integers x and y such that $36x + 28y = \gcd(36, 28)$.
12. (*) Prove *Pascal's Identity*. (Proposition 8.10 in the lecture notes.)
13. (*) Remember that the Fibonacci numbers are defined by the following facts: $F(0) = 0$, $F(1) = 1$, and $F(n) = F(n-1) + F(n-2)$ when $n \geq 2$.
   (a) What is the remainder of $F(n)$ divided by $F(n-1)$ (when $n \geq 2$)?
   (b) Use the previous fact, Proposition 9.14, and induction, to prove that $\gcd(F(n), F(n-1)) = 1$ when $n \geq 2$.

14. (*) Let a and b be positive integers. Prove that

$$\log_b n = \frac{1}{\log_a b} \log_a n$$

for any positive number n.